ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN COMPUTER SCIENCE/ IT

CCS3222 FREE AND OPEN SOURCE SYSTEMS

FINAL EXAMINATION

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 27th NOVEMBER, 2023.

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

Study the description below and answer the questions that follow:

QUESTION 1

Software that is licensed for unrestricted use, modification, and distribution is known as Free Open-Source Software (FOSS), sometimes referred to as just Open Source or Free Software. A sort of legal Golden Rule that applies to most FOSS licenses mandates that all modifications, including fixes and upgrades, must be provided under the same license. This builds the crucial trust between users and developers, which in turn fosters the growth of big, enduring communities that eventually expand the capabilities of the software. The portion of software that most users never see is called source code; programmers use it to modify the behavior of programs and applications. A program can be altered, expanded upon, or have malfunctioning sections fixed by programmers with access to the source code. OSS usually comes with a license that lets programmers alter the program to suit their needs and restricts how it may be shared.

For several decades, FOSS has been improving gradually. It powers most of the Internet, offers greater functionality, security, and sustainability than commercial software at a significantly lower cost, and is becoming more and more popular as the go-to software choice for both individuals and businesses. The idea of making source code freely available originated in 1983 from an ideological movement informally founded by Richard Stallman, a programmer at Massachusetts Institute of Technology (MIT). Stallman believed that software should be accessible to programmers so they could modify it as they wished, with the goal of understanding it, learning about it, and improving it. Stallman began releasing free code under his own license, called the GNU Public License.

Answer the following questions:

a) Discuss the concept of "Free open source software (FOSS)" (4 marks)

b) The idea of making source code freely available originated in 1983 from an ideological movement informally founded by Richard Stallman, a programmer at Massachusetts Institute of Technology (MIT). Discuss three (3) important reasons of making source code freely available. (6 marks)

c) Outline two (2) differences between FOSS and commercial software (4 marks)

d) Explain four (4) advantages of using FOSS in an organization. (8 marks)

e) Explain four (4) disadvantages of using FOSS in an organization. (8 marks)

f) Certain organization choose to purchase software, usually called "**off the shelf software**", discuss three (3) advantages of purchasing such software. (6 marks)

g) Depending on the size of the business organization and nature of operations, it is important to hire programmers. Explain the importance of this. (4 marks)

[Total: 40 marks]

SECTION B: Answer any three (3) questions in this section.

QUESTION 2

a) In recent past, there has been many open-source license types under which hundreds of open-source software have been published. Each of these licenses has its own terms and conditions of usage. Describe three (3) examples of FOSS licenses. (6 marks)

b) The 'free' in Free and Open-Source Software (FOSS) doesn't mean it is free of cost. It means freedom to run, modify and distribute the software. Discuss two (2) costs that may come with running FOSS in an Organization. (4 marks)

c) Data backup is the process of storing additional copies of your data in physical or virtual locations distinct from your data files in storage. With the use of one (1) practical example in each case, explain the following types of backups:

i) Full backup (4 marks)

ii) Differential backup (3 marks)

iii) Incremental backup (3 marks)

[Total: 20 marks]

QUESTION 3

a) Linux, being an open-source software also means that it has security risks and vulnerabilities. An organization needs to know how to secure the Linux server both to protect the business and the data for customers. Discuss any three (3) security measures. (6 marks)

b) According to Richard Stallman's Free Software definition, "free software" requires the software's licensing to respect the civil liberties and human rights of four essential freedoms. Discuss these freedoms. (8 marks)

c) With an understanding of what open-source licenses are and their benefits, it's also useful to know what are the main two categories of these licenses, the different types, and their requirements. Differentiate between **Copyleft** and **Permissive Licenses**. (6 marks)

[Total: 20 marks]

QUESTION 4

a) Describe any three (3) applications of python in industry today. (3 marks)

b) Variables in python can be declared as shown below. Explain the meaning and output of the print statements for x, y and z.

```
x = 5
y = "John"
z = 22.7
print(type(x))
print(type(y))
print(type(z))
```

(1 mark)
(1 mark)
(1 mark)

c) State the output of the following program

```
x = "Sunday, the first day of the week!"

print(a[4:9])
```

(2 marks)

d) Write a simple python program to concatenate two strings. (4 marks)

e) Python uses many operators, some of them are listed below. With the use one (1) example in each case, explain the ones listed below:

i) Arithmetic operators (2 marks)
ii) Assignment operators (2 marks)
iii) Comparison operators (2 marks)
iv) Logical operators (2 marks)

[Total: 20 marks]

QUESTION 5

a) Microsoft Windows and Linux are regarded as one of the most famous operating systems used by business organizations. Discuss three (3) differences between these operating systems. (6 marks)

b) Kali Linux is an open-source, Debian-based Linux distribution geared towards various information security tasks. Discuss any two (2) embedded in Kali Linux. (4 marks)

c) In Python, Lists and Tuples are used to store multiple items in a single variable. Discuss the difference between them. (4 marks)

d) Explain any three (3) consequences of running pirated software in a business organization. (6 marks)

[Total: 20 marks]